Answer **all** the questions in the spaces provided.

1 **(a)** A speed boat has an initial velocity of 12 m s^{-1} south. The speed boat then changes its course to a final velocity of 16 m s^{-1} east. Determine the magnitude of the change in velocity, Δv, of the speed boat, and the angle between vector Δv and the north direction.

magnitude of Δv = …….. m s^{-1} [1]

angle between Δv and north = …….° [1]

(b) A student uses a pair of vernier caliper to take several measurements of the diameter of a rubber ball. He applies different pressures when closing the gap of the vernier caliper. State and explain whether this introduces a systematic error or random error into the readings.

…….....

…….....

…….....[2]

(c) The energy per unit time, P radiated by an object with a surface area A at thermodynamic temperature T is given by

$$P = e\sigma A T^4$$

where e is the emissivity of the surface and σ is the Stefan-Boltzmann constant.

(i) Given that the SI unit for σ is W m^{-2} K^{-4}, determine the base units of emissivity, e.

base units of e = ... [2]

(ii) In an experiment to determine emissivity e of a circular surface area of diameter d of an object, the following measurements are taken:

$P = (3.0 \pm 0.2)$ W
$d = (5.0 \pm 0.1)$ cm
$T = (500 \pm 1)$ K
$\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}

Determine the value of emissivity, e of the surface and express it with its associated uncertainty.

e = ... $\pm$ .... [4]